Mass-balance dosimetry for in vitro PFAS testing: a source-verified reassessment of two models against three uptake datasets

Working manuscript, 22 October 2026. Analysis, figures and tables are reproducible from `tk_review/scripts/`; see *Data and code availability* for the script behind each section and figure. Four figures and ten tables. All model parameters and all exposure conditions are verified against primary sources; the provenance of every reference is stated after the reference list.

Abstract

Background. Quantitative in vitro to in vivo extrapolation (QIVIVE) for per- and polyfluoroalkyl substances (PFAS) requires knowing how much of an administered dose actually reaches cells in a bioassay. Nominal medium concentrations overstate it, sometimes by three orders of magnitude, because PFAS bind extensively to serum albumin and sorb to vessel walls. Equilibrium mass-balance models (IV-MBM) were developed to recover the freely dissolved and cellular concentrations from first principles.

Objective. To reassess two published IV-MBM formulations — the single-descriptor IV-MBM EQP v2.0 of Armitage et al. and the concentration-dependent balance of Qin et al. — against the three published datasets reporting PFAS cellular uptake in human liver cell lines, with all model parameters and all exposure conditions verified against primary sources.

Approach. Both models were re-implemented from their published equations and supplementary parameterisations. The three uptake datasets (Rosenmai et al.; Amstutz et al.; Fragki et al.) were harmonised onto the fraction-of-dose axis. Exposure conditions were read from the primary methods sections rather than inherited from derived descriptions. Model residuals were decomposed by study, compound, dose and chain length, and each candidate parameter was replaced with the best available independent measurement.

Principal findings. (i) Reading the primary methods revealed that one of the three assays was dosed in 2% rather than 10% serum, a five-fold error in the dominant sorptive phase that is not apparent from secondary descriptions; correcting it degraded the apparent fit of the dataset that had previously looked best. (ii) On corrected media both models over-predict cellular uptake — geometric-mean bias 3.3× (Armitage) and 6.4× (Qin) over 29 observations. (iii) Every substitution of a best-available measured parameter made the fit worse, in each case because the measurement moved the balance in the direction the residual did not need. (iv) The residual separates into a level error and a chain-length error with different causes, and the separation is clean: the chain-length behaviour is a property of the partition coefficients and the level is a property of the volumes and compositions, each invariant to changes in the other. Reconstructing a third published formulation — the measured-parameter linear balance of Fischer et al., using measured PFAS albumin and membrane coefficients — removes the chain-length drift entirely over nCF 5–7 while leaving a 4.6× over-prediction, so no parameterisation tested here corrects the level. The over-steepening of the lipid term that causes the drift is corroborated by two independent laboratories, two lipid preparations and two analytical routes (+0.36 to +0.48 decades per CF_2 measured, against +0.78 predicted). (v) The same formalism applied to human serum reproduces measured whole-serum partitioning for 26 PFAS with no fitted parameter, establishing that the accounting is sound and localising the failure to the in vitro system. (vi) All three assays operate 5,500–27,700× above the physiological ligand:protein ratio, confirming and quantifying the regime concern raised by Qin et al.

Implication. Improved bookkeeping within the well yields a better description of the experiment but does not by itself yield a better oral equivalent dose, because the experiments occupy a binding regime that does not occur in vivo. We give specific reporting requirements that would make such assays usable for QIVIVE.

Keywords: PFAS; in vitro dosimetry; mass balance; QIVIVE; serum albumin; ligand:protein ratio; freely dissolved concentration

1. Introduction

1.1 The nominal concentration problem

In vitro bioactivity data are reported almost universally on a nominal basis: the concentration weighed into the medium. For chemicals that distribute appreciably out of the aqueous phase, that number is not the exposure the cells experienced. Chemical partitions into serum lipids and proteins, sorbs to the plastic of the well, is taken up by the cells themselves, and — for volatiles — escapes to the headspace. The freely dissolved concentration that drives uptake can be orders of magnitude below nominal [1,12].

This matters because the purpose of the measurement is extrapolation. If an effect concentration measured in a dish is to be compared with a blood concentration in a person, the two must be expressed on comparable bases. A bioassay medium may contain anywhere from 0 to 20% serum; blood is blood. Using a nominal effect concentration as a direct surrogate for a blood concentration embeds a bias whose size depends on hydrophobicity and on assay composition [1].

Equilibrium mass-balance models were developed to close this gap [1,2,12,13,16]. They are now sufficiently mature that several independent implementations exist, and a comparison of their features has been published [17, cited via 1].

1.2 Why PFAS are a difficult case

The spLFER-based mass-balance approach was calibrated on neutral, hydrophobic organics, for which a single descriptor — the octanol–water partition ratio — predicts partitioning to most biological phases well. PFAS violate the premises of that approach in three distinct ways.

First, they are surfactants: a perfluorinated tail with an ionisable head group. Their behaviour at interfaces and in mixed phases is not well captured by a bulk hydrophobicity descriptor, and the descriptor itself is difficult to measure for them.

Second, they are essentially fully anionic at physiological pH. Ionisation is handled in these models by scaling the neutral-form partition ratio by a phase-specific penalty. For serum albumin, Armitage et al. set that penalty to zero for acids — the anion is assumed to bind albumin exactly as well as the neutral form would (ref. [1], SI Table S1). This is a modelling decision rather than a measurement, and it is consequential: it is a major reason albumin ends up carrying the great majority of the chemical in these calculations.

Third, PFAS bind albumin at discrete sites rather than dissolving into it diffusely. Site binding saturates. Qin et al. [3] measured PFAS–albumin isotherms over five orders of magnitude and found the binding strongly concentration-dependent, which is the basis of their model and the reason the linearity assumption common to spLFER approaches cannot simply be carried over.

1.3 Aims

This reassessment addresses four questions:

1. How well do the two models predict PFAS cellular uptake when both are implemented from their own published parameterisations?
2. Are the available uptake data mutually consistent enough to adjudicate between models?
3. Where does the residual lie, and can it be attributed to a parameter?
4. What do the answers imply for the use of these assays in QIVIVE?

We stress at the outset that question 4 was raised, and answered in principle, by Qin et al. [3], who hypothesised that concentration-dependent protein binding causes the binding regime in bioassays to differ substantially from that in plasma, with consequences for QIVIVE. What follows quantifies that gap for the specific uptake datasets used to parameterise and test these models, and examines what it leaves achievable.

2. The mass-balance formalism

2.1 Governing relation

The well is treated as a set of phases at equilibrium. Conservation of the administered amount, with each phase concentration expressed relative to the freely dissolved concentration by a distribution ratio D, gives

$$n_{total} = C_{free} \cdot (V_w + \Sigma_i D_i V_i)$$

so that $C_{free} = n_{total} / S$, where $S \equiv V_w + \Sigma D_i V_i$ is the total sorptive capacity of the well in units of water-equivalent volume. The fraction of the administered dose residing in cells follows as $x_{cell} = D_{cell} V_{cell} / S$.

Models differ only in how the D_i are obtained. The accounting is not in dispute.

2.2 Assumptions

The derivation requires: equilibrium; no degradation, metabolism or active transport; no losses; additive, non-interacting phases; and — the load-bearing one — linear sorption, i.e. constant activity coefficients, which holds only while the chemical is dilute with respect to every sorptive phase.

For PFAS in these assays, equilibrium is defensible: Fragki et al. [7] report cellular steady state within approximately 1 h against a 24 h exposure. Metabolic stability is defensible for perfluoroalkyl acids, though not for fluorotelomer alcohols. Active transport is assumed absent and cannot be checked. The linearity assumption is the one that fails outright, and §7 quantifies by how much.

2.3 Choice of output metric

Two metrics are available: the cellular concentration and the fraction of the dose in cells. The published uptake data are reported as amount per mass of cell protein, so converting them to a concentration requires a cell volume, while converting them to a fraction requires only protein per cell.

It should be noted — because it is easy to get wrong — that this does *not* make the fraction axis immune to cell-volume uncertainty. The predicted fraction is proportional to cell volume even though the measured fraction is not, with the result that the ratio of predicted to measured scales with assumed cell volume on *either* axis. What the fraction axis does buy is that the three datasets' *measurements* become mutually comparable without any cell volume being assumed. We use it for that reason.

3. The two models

3.1 IV-MBM EQP v2.0 (Armitage et al.)

The model predicts all required distribution ratios from the neutral-form octanol–water partition ratio via sorbent-phase-specific linear free energy relationships (spLFERs), with ionisation handled by phase-specific scaling factors [1]. The cell is represented as a proximate composition — storage lipid, membrane lipid, bulk protein and water — each contributing in proportion to its volume fraction. Plastic sorption is included via one of two published spLFERs.

We verified every parameter used here against the published supplementary material: both spLFERs (SI §S1.3), all four ionisation scaling factors (SI Table S1), and the cell-line compositions (SI Table S5). All agree exactly.

One parameter could not be verified. How bulk cell protein is converted into sorptive capacity is stated nowhere in the main text or the supplementary material; §S1 covers anionic phospholipids, lysosomal sequestration and the spLFERs, then defers to ref. [2] for temperature and ionic-strength adjustments only. The earlier paper does not contain it either, and the reason is stronger than the absence of a documented value: in v1.0 there is no protein term at all. Its mass balance represents the cell as a single lumped phase, $K_{CW} = K_{OW} \cdot f_L$, where f_L is "the volume fraction of lipid equivalents (i.e., pseudo-octanol content)" after the standard multimedia-model convention, assumed to be 0.05 for cells and varied over 0.01–0.05 [2]. There is no membrane-lipid phase and no cell-water phase either. The proximate composition of v2.0 is a new construction, so the coefficient attached to its protein phase cannot be recovered from the earlier record; and the 3.5% convention does not appear there. The implementation used here therefore adopts the convention in EPA's `httk` package, under which 3.5% of bulk protein is treated as octanol-equivalent.

Two further points follow from the earlier paper and bear on how much this matters.

First, **v1.0 excludes PFAS by its own statement**: it "is intended for neutral organic chemicals and should only be applied to ionogenic organic chemicals (IOCs) if the compound is predominantly neutral at the test pH", and its future-work list names the gap — "to expand the approach for neutral organic chemicals to ionogenic organic chemicals that are predominantly charged in the test system" [2]. The IOC machinery of v2.0 is that extension. Fully anionic PFAS are among the first applications of it, not a case on which the formalism was calibrated.

Second, the undocumented coefficient has **little leverage at its own value**. With protein contributing 0.05–0.25% of the cell's sorptive capacity, the modelled cell is 79% water for the shortest chain tested and 99.7% membrane lipid for the longest: the cell is, in effect, a lipid droplet. The model's output is therefore insensitive to the value of 0.035 over any plausible range. What has leverage is the *choice* it encodes — octanol-equivalence for bulk protein — and replacing that choice with a measured structural-protein affinity enlarges the cell sink by 1.4–3.4× (§6.4). The distinction matters for reading what follows: results that turn on the modelling choice are robust; results that would turn on the particular constant do not arise, because none is sensitive to it.

3.2 The concentration-dependent balance (Qin et al.)

Qin et al. [3] measured PFAS binding to bovine serum albumin, lipid, human plasma, assay medium and cells by C18 solid-phase microextraction over five orders of magnitude, and represent albumin binding as a saturable term plus a linear non-specific term. Their biomaterial–water ratio is

$$D_{i/w} = D_{BSA/w}\cdot(V_{prot}/V_{prot+lip}) + D_{lip/w}\cdot(V_{lip}/V_{prot+lip})$$

and the bioassay balance is

$$C_{free} = n_{tot} / [(D_{cell/w}\cdot V_{cell,prot+lip} + V_{cell,w}) + (D_{medium/w}\cdot V_{medium,prot+lip} + V_{medium,w})]$$

Two features of this form are easy to misread and are worth stating explicitly, because both bear on the comparison in §3.3. First, the normalising volume is the *sorptive* volume $V_{prot+lip}$, not the volume of the biomaterial; the authors give the units of their tabulated cell–water ratio as $L_{w}/L_{cell,prot+lip}$, confirming it. Second, cell water does appear in the balance, but as a non-sorbing phase at D = 1 pooled with the medium water rather than inside $D_{cell/w}$. The distinction the authors draw against the earlier Henneberger formulation is about the *numerator*: they "considered only proteins and lipids of the cell homogenate as the main sorptive phase", so cell water lowers the free concentration but does not count as PFAS in the cell. For the cell term itself the composition is protein and lipid only, 95% protein by sorptive volume for their HEK293H line (8.14% protein and 0.432% lipid of cell volume; water 91.6%).

The protein term is the measured non-specific protein–water ratio rather than an octanol surrogate: the authors take $D_{non\text{-}specific}$ "as a surrogate for partitioning to structural proteins", on the argument that cytoskeletal protein binds organic acids more weakly than albumin does.

The saturable albumin term can be derived from mass action: for n independent sites, the site concentration within pure albumin is $n\cdot rho/M$. With the authors' own stated values (rho = 1.36 kg/L, M = 66,463 Da) this evaluates to n × 20,462 µM, which reproduces their published site-capacity column to within 0.2% across all four measured compounds and implies n = 0.45 for PFBA and 1.24 for PFOA. In the trace limit the term collapses to a constant, recovering the standard bridge between a binding constant and a partition coefficient.

Parameters were measured for four compounds (PFBA, PFOA, PFHxS, PFOS). Values for PFHxA and PFHpA are interpolated between the two carboxylate anchors and PFNA is extrapolated beyond both. Only the lipid term has a published interpolation rule — a regression of log $D_{lip/w}$ on the number of fluorinated carbons with slope +0.476 and intercept +0.172, which the authors used to supply the PFBA value and which reproduces their tabulated carboxylate entries exactly.

3.3 Structural comparison

Table 1. Structural differences between the two models.

	IV-MBM EQP v2.0	**Qin et al.**
source of *D*	predicted from log KOW	measured, then interpolated
albumin	linear, piecewise spLFER	saturable (site binding + non-specific)

	IV-MBM EQP v2.0	Qin et al.
albumin anion penalty	0.0 (acids)	not applicable; measured on the anion
cell phases	storage lipid, membrane lipid, 3.5%·protein, water	protein, lipid; water non-sorbing
cell composition at short chain	~79% water	~95% protein (sorptive volume)
cell composition at long chain	~99.7% membrane lipid	~95% protein (sorptive volume)
dose dependence	linear by construction	non-linear
chemical coverage	any, from one descriptor	four measured compounds

The two models hold opposite views of what a cell is. Their predicted uptake nonetheless agrees to within a factor of ~2.5 in many cases, because albumin dominates the denominator in both and masks the difference. Agreement between them should not be read as corroboration of either cell representation.

3.4 The wider landscape, and why two models

Proença et al. [17] catalogue fourteen published in vitro distribution models and classify them by the strategy used to obtain partitioning data: experimental values for specific chemicals, experimental values for a specific system, or QSPRs. Ten of the fourteen are specific to particular chemicals or particular setups; four are generic. The models relevant here are the generic ones, and they differ along axes that matter for PFAS:

Table 1b. Design axes across published in vitro distribution approaches.

	source of partitioning	albumin treatment	cell representation	normalising volume
Armitage et al. v1.0 [2]	KOW × lipid-equivalent fraction	linear	one lumped pseudo-octanol phase	Vcell
Armitage et al. v2.0 [1]	spLFERs from log KOW	linear, piecewise	lipid + 3.5%·protein + water	Vcell
Fischer et al. [12]	**measured**, else PP-LFER / 3D-QSAR / COSMOmic	linear	protein + lipid + water, measured; DBSA/w for all protein	Vcell
Qin et al. [3]	measured, PFAS-specific	**saturable**	protein + lipid; water non-sorbing	Vprot+lip
Stadnicka-Michalak et al. [18]	regressions on log KOW	linear	kinetic, not equilibrium	—
VCBA / PBK-coupled [19]	varies	linear	kinetic, coupled to PBK	—

Two of these — the kinetic approaches — answer a different question, since the PFAS uptake data here reach steady state within ~1 h against a 24 h exposure and an equilibrium treatment is adequate. Of the equilibrium models, the two tested in full occupy opposite corners of the first two axes, which is why they were selected. The third corner — measured biopartitioning with a *linear* albumin term — is the Fischer et al. design, and §6.7 reconstructs it from published PFAS measurements without reimplementing anyone's code.

Proença et al. also ran five of the generic models side by side on six test chemicals in HepaRG cells at 1 µM, with and without 10% FBS. Two of their observations bear directly on the results below. The Fischer et al. model predicts the highest cellular concentration of the five, and the Armitage model predicts "substantially lower" values, which they attribute to "both the cellular compartments added (i.e. just lipid or protein and lipid) and QSPRs used (log KOW-based QSPRs or PP-LFERs based on other descriptors)". That ordering and that attribution are reproduced independently in §6.7.2 below. They also state the limit of their own exercise: "A comparison of model predictive performance is only possible with adequate experimental data, and is beyond the scope of this review" — and, in their discussion, that "especially for the generic models more extensive experimental validation is long overdue". The present work is an attempt at that validation for one chemical class.

Three further statements in that review anticipate findings reported below, and are credited here rather than claimed. On albumin: "for anionic chemicals sorption to albumin cannot be predicted with a simple partition since saturation of albumin binding sites for these chemicals occurs at concentrations relevant for toxicity testing", citing Henneberger et al. 2020 [21] — the argument of §7, which predates Qin et al. and which they sharpen by noting that organic acids "appear to only have one binding site, which results in the saturation at relatively low concentrations". On lipid: "a bulk solvent such as octanol is not a good surrogate for anisotropic lipid bilayer membranes, especially for partitioning of ionised

chemicals", and membrane partitioning "is more similar between neutral and charged species than KOW based predictions will show" — the mechanism behind §6.7's central result. On the applicability domain: "the main limitation remains accurate QSPRs to predict partitioning of heavily charged chemicals". PFAS at pH 7.4 are heavily charged, so the compounds modelled here sit on the boundary of the domain the review identifies, which is the reason for testing rather than assuming.

Two details in the last column of Table 1b are easy to lose and both caused errors in earlier drafts of this work. The three equilibrium balances do not report cell–water distribution on a common basis: Armitage and Fischer normalise by total cell volume, Qin by the volume of cell protein plus lipid, and the two differ by a factor of ~17 for a hepatic composition. And the treatment of cell protein is not a detail of parameterisation but a structural choice: Armitage gives it an octanol equivalence, which makes it negligible; Fischer gives it the albumin surrogate, which makes it comparable to the lipid term; Qin gives it a measured non-specific protein coefficient, which makes it dominant. A model's cell is defined as much by that choice as by its composition.

4. The empirical basis

4.1 Three datasets

Three studies report PFAS cellular uptake in human liver cell lines on a fraction-of-dose basis: Rosenmai et al. [5] (HepG2, 14 PFAS, three doses), Amstutz et al. [6] (HepG2, 13 PFAS, one dose) and Fragki et al. [7] (differentiated HepaRG, four PFAS, five doses). Twenty-nine observations overlap the five perfluoroalkyl carboxylates for which both models can be parameterised, and those 29 are the basis of §§6.1–6.8. Fragki et al. also measured two sulfonates, PFHxS and PFOS; those ten observations are analysed separately in §6.9, because pooling them would confound head group with study.

Two provenance notes on the third dataset. First, its biokinetic measurements are in Fragki et al. [7]; the companion paper from the same group on the same cell system, Louisse et al. [8], reports hepatotoxic potencies and gene-expression changes but no biokinetics, and is sometimes cited for the uptake data. All uptake values used here are from ref. [7], and the exposure conditions in Table 2 are from its methods section, not from ref. [8].

Second, ref. [7] reports its uptake two ways: as a concentration series in nmol per mg cell protein, and as the **fraction of the dose taken up** (its Table 3). An earlier version of this work used the first and converted it with an assumed protein content per well, which is not stated in the paper. The paper's own two numbers pin that content instead — protein = uptake fraction × n_{total} ÷ (nmol/mg) gives 0.30–0.34 mg per well across the PFOA series, a coefficient of variation of 6% — and that is 2.1× the assumed value, implying roughly 4.8 × 105 differentiated cells per 24-well rather than the 2.2 × 105 assumed. The reported fraction needs no protein assumption at all and is exactly the quantity both models predict, so it is used here. The correction lowers the Fragki residual from 10.80× to 6.66× (Armitage) and 14.92× to 9.19× (Qin), and the pooled bias from 3.87× to 3.28× and 7.52× to 6.36×. **Every result below uses the directly reported fractions.** It is worth stating plainly that this was our error, not the source's: the paper reports the quantity needed, and an avoidable conversion was interposed in front of it.

4.2 Exposure conditions, read from source

Table 2. Exposure conditions as stated in the primary methods sections. Entries in bold differ from what a reader would infer from the cell-culture section alone.

	Rosenmai [5]	**Amstutz [6]**	**Fragki [7]**
cell line	HepG2	HepG2	differentiated HepaRG
growth medium serum	10% FBS	10% FBS	10% FBS
exposure medium serum	10% FBS	10% FBS	**2% FBS**
co-solvent	**3% ethanol**	0.1% DMSO	0.5% DMSO
plate format	12-well, 3 wells pooled	96-well	24-well
volume dosed -> after t=0 sample	not stated	250 -> 200 µL	550 -> 500 µL

	Rosenmai [5]	Amstutz [6]	Fragki [7]
wash	5× ice-cold PBS	5× DPBS	5× DPBS
uptake normalised to	nominal	**recovered**	nominal

Two entries require comment. Fragki et al. state that exposure was conducted in "assay medium (growth medium containing 2% FBS)"; the 10% figure belongs to the growth and differentiation medium. Because serum fraction multiplies the volume of the dominant sorptive phase, this is a five-fold difference in the parameter to which the balance is most sensitive, and it is not recoverable from descriptions that quote the culture conditions. Second, Amstutz et al. normalise their intracellular fraction to *recovered* rather than administered mass, so their denominator excludes material lost to the vessel; their values are not strictly on the same axis as the other two.

Rosenmai et al. do not state a working volume; 1 mL per well is assumed, as in all prior analyses of this dataset known to us.

4.3 Inter-study consistency

The three datasets disagree with one another before any model is involved. At 100 µM, cellular uptake of PFNA is reported as 0.31% [5], 0.30% [6] and 4.39% [7] — a 14-fold spread. For PFHxA at the same dose the two HepG2 studies differ by 3.6-fold (0.14% [5] vs 0.039% [6]) despite using the same cell line and nominally similar protocols; part of this is attributable to the different normalisation noted above.

Any model-performance statement must be read against this. A model disagreeing with these data by a factor of four is disagreeing with a target that is itself uncertain by a comparable factor.

5. Methods of this reassessment

Both models were implemented in Python from their published equations and parameterisations (`mbm_published_equations.py`). Armitage parameters were taken from ref. [1] and its supplementary material; Qin parameters from ref. [3]. Where the Armitage implementation required a parameter not present in the published record (cell protein sorption, §3.1), the `httk` convention was used and is flagged throughout.

Assay geometry, cell number, serum fraction and dose series were taken from the primary publications (Table 2). Measured uptake was placed on the fraction-of-nominal-dose axis. Model performance is reported as geometric-mean bias (predicted ÷ measured) and geometric-mean absolute error.

Residuals were decomposed by study, by compound, by dose and against perfluorocarbon chain length. Four parameters were then replaced in turn with the best available independent measurement: albumin partitioning from Fischer et al. [9]; cell partitioning from the measured values in ref. [3]; cell protein sorption from ref. [4]; and cell volume and composition from Fischer et al.'s measurements on HepG2 [12]. A third formulation, the measured-parameter linear balance of Fischer et al. [12], was then reconstructed from its published equations and run on the same data with measured PFAS coefficients throughout.

6. Results

6.1 Effect of the medium correction

Implementing Fragki et al. at the serum fraction stated in their methods rather than the growth-medium value changes that dataset's geometric-mean bias from 0.95× to 9.20× (Qin) and from 1.53× to 6.66× (Armitage). Pooled across all 29 observations, bias moves from 2.91× to 6.36× (Qin) and 1.97× to 3.28× (Armitage).

The direction is worth stating plainly: the dataset that had appeared to be the models' best case was the one being supplied with five times more albumin than it contained. Correcting an error degraded the apparent fit. This is the expected behaviour when a model is accidentally compensated, and it is a cautionary result for any benchmarking exercise that inherits assay conditions from secondary sources.

The Armitage response (4.4×) matches the denominator arithmetic, since its albumin term is linear. The Qin response (9.7×) exceeds it, because a saturable sink amplifies an error in its own size: reducing albumin raises the free concentration, which raises site occupancy, which lowers the apparent distribution ratio, which reduces capacity further. The amplification is largest at intermediate doses, where the 10% and 2% wells fall on opposite sides of saturation, and vanishes at the top dose where both are saturated.

6.2 Overall performance

Table 3. Model performance on corrected media, 29 observations.

	geometric-mean bias	**geometric-mean **	**error**
IV-MBM EQP v2.0	**3.28×**	**3.41×**	
Qin et al.	6.36×	6.36×	

by study	**Armitage**	**Qin**
Rosenmai [5] (n=15)	1.97×	3.72×
Amstutz [6] (n=4)	3.77×	18.95×
Fragki [7] (n=10)	6.66×	9.19×

Both models over-predict throughout. The error is not uniform across compounds: for PFBA–PFHpA, which only the two 10% FBS studies measure, Armitage bias is 1.2–1.5×, while PFOA and PFNA — the compounds Fragki contributes — carry almost the entire residual.

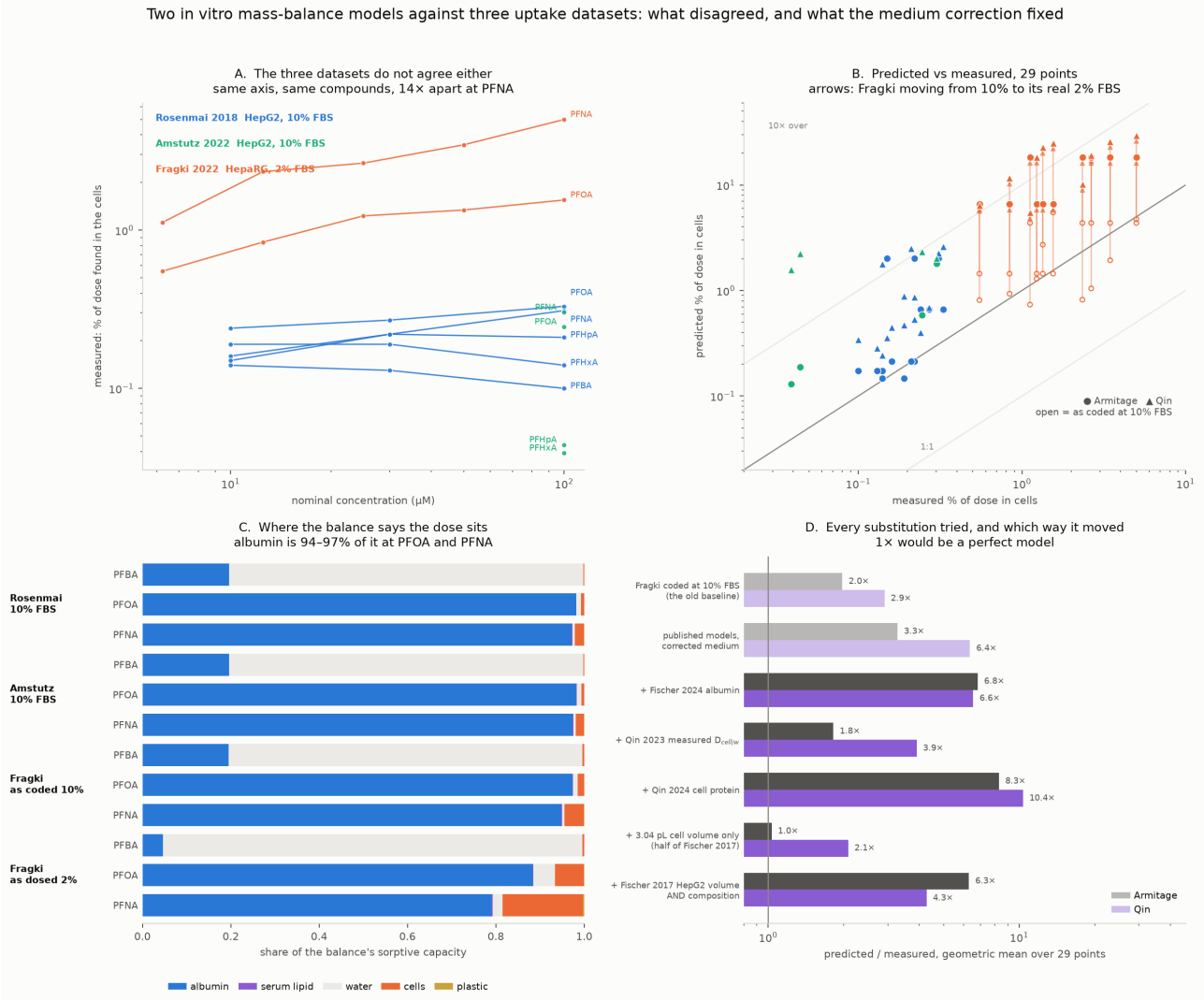


Figure 1. A, the three uptake datasets on a common axis: measured fraction of dose in cells against nominal concentration, for the five compounds both models can be parameterised for. The datasets disagree with each other by up to 14× at PFNA, so no single model can match all three. **B**, predicted against measured for the 29 observations; open symbols are Fragki et al. as coded at 10% FBS and filled symbols after the correction to the 2% at which the assay was actually dosed (§6.1), with arrows joining the pairs. The grey diagonals are the 1:1 line and 10× over. **C**, where the balance puts the dose in each well, by sorptive phase. Albumin carries 94–97% of the capacity at PFOA and PFNA in the 10% FBS assays; the cells carry a visible share only in Fragki's 2% FBS medium at the long-chain end. **D**, the geometric-mean bias of each parameter substitution, with 1× a perfect model (§6.1, §6.4). Every best-available measurement moves at least one model further from the data. The last two bars are the same substitution taken two ways: Fischer et al.'s HepG2 cell volume alone, and that volume together with the cell composition they measured in the same table.

6.3 Dose dependence

Armitage is linear by construction and its fitted dose exponent is pinned at 1.00. Measured exponents run 0.85–1.42 and rise with chain length. Qin reproduces that ordering, overshooting by 0.30–0.71 on the Rosenmai series and by 0.07–0.20 on the Fragki series. This is the one respect in which the additional machinery of the saturable model buys something the linear model structurally cannot deliver, and it is also the one respect in which the medium correction *improves* Qin: the Fragki exponents fall from 1.71 and 1.66 to 1.49 and 1.62 against a measured 1.42, indicating that part of the curvature previously attributed to chemistry was an artefact of an oversized albumin sink.

6.4 Parameter substitutions

Table 4. Effect of replacing each parameter with the best available independent measurement. Bias is the geometric mean over 29 observations.

substitution	direction of the measurement	Armitage	Qin
published models	—	3.28×	6.36×

substitution	direction of the measurement	Armitage	Qin
albumin partitioning [9]	**below** both models	6.84×	6.56×
cell partitioning (measured, [3])	mixed	1.82×	3.90×
cell protein sorption [4]	**above** both models	8.30×	10.36×
cell volume alone (measured HepG2 [12])	—	1.03×	2.08×
cell volume **and** composition [12]	—	6.30×	4.27×

Each measurement moves the balance in the direction the residual does not require. A weaker albumin reduces competition and sends more chemical into the cell; a stronger cell-protein affinity enlarges the cell sink and does the same. Both models already over-predict.

The cell-partitioning substitution improves the aggregate but at the cost of destroying the chain-length trend, over-predicting PFBA by 19.7× (Qin) and 23.6× (Armitage) while landing within a factor of 1.4 at PFNA. Aggregate improvement achieved by averaging a large miss at one end of a series against a good fit at the other is not evidence of a better parameter.

The cell-volume row requires care, and an earlier version of this work read it wrongly. Taken alone it looks like much the best fit in the study — 3.28× to 1.03×, essentially unbiased — and it is also the only parameter that rescales every compound by the same factor, which is what a chain-length-independent offset requires. But the measured cell volume comes from Fischer et al. [12], who measured the *composition* of the same HepG2 cells in the same table. Using the volume while retaining Armitage's composition takes half of one measurement. Taking it whole moves the bias to 6.30× — the measured hepatic lipid fraction is 6.9× larger than the published parameterisation's while the cell volume is only 3.3× smaller, so the cell sink roughly doubles.

Every best-available measurement therefore worsens the fit, with no exception. This is the central negative result of the reassessment: a parameter can be the most influential term in a model and not be the term that is wrong. Albumin carries 94–97% of the sorptive capacity for the longer chains, which makes it the obvious suspect and the obvious thing to tune; the best available measurement of it moves the prediction further from the data. The corollary is methodological: a substitution that improves a fit should be checked for whether it imported only the convenient half of its source.

6.5 Chain-length structure of the residual

Armitage's over-prediction rises monotonically with perfluorocarbon chain length; Qin's does not. Within the Rosenmai dataset alone — one laboratory, one cell line, one medium, one vehicle — the Armitage residual climbs 0.86× to 9.33× between PFHxA and PFNA. Qin shows no trend within any single study.

The cause is not either distribution ratio alone. Where albumin dominates the denominator, the predicted fraction in cells varies as the *ratio* $D_{cell}/D_{albumin}$, so the residual's chain-length slope is the error in the slope of that ratio.

Table 5. Chain-length dependence over nCF 5–8, decades per CF_2.

	albumin	cell	ratio	residual, observed
IV-MBM EQP v2.0	+0.386	+0.769	**+0.383**	**+0.345**
Qin et al.	+0.547	+0.515	**-0.032**	**-0.074**
measured albumin [9]	**+0.540**	—	—	—

The ratio slope predicts the residual slope to within the uncertainty of a four-point fit. Armitage's albumin spLFER rises more slowly with chain length than the measurement, because the piecewise form flattens above log K_{OW} 4.5 — a feature calibrated on neutral organics and pharmaceuticals, where a binding cavity of finite depth does plateau. Its cell term rises faster than a constant ratio would require, because a cell represented as lipid gains affinity with chain length faster than albumin does. The two errors compound, contributing roughly 40% and 60% of the gap respectively.

Qin's albumin slope sits within 0.007 of the measured value and his ratio is flat. **The two models therefore fail in different ways: Qin has the chain-length behaviour approximately right and the level wrong; Armitage has the level closer and the chain-length behaviour wrong.**

Independent estimates of the cell distribution ratio's chain-length slope span +0.177 (cell homogenate, [3]) to +0.782 (the Armitage parameterisation), with the uptake-implied value (+0.407) and whole-tissue unbound fractions [11] (+0.46 to +0.53) in between. The quantity is bounded, not known.

6.6 What does not explain the residual

Two candidate artefacts were tested and rejected.

Wash losses. All three studies wash five times with cold buffer, so it cannot be a between-study variable; and the direction is wrong, since washing should preferentially strip the shortest, most weakly bound chains, whereas over-prediction rises with chain length.

Cytotoxicity at the long-chain end. Reduced protein content at high dose is documented in two of the three studies, and Fragki et al. excluded PFNA at 100 µM from their own analysis on those grounds. If cell loss were suppressing measured uptake, the chain-length trend would be strongest at the highest dose. It is strongest at the lowest (+0.194 at 10 µM, +0.099 at 100 µM).

Albumin secreted by the cells. Two of the three datasets use hepatic lines, which synthesise albumin, so the albumin sink could in principle exceed what the serum supplement accounts for. Proença et al. [17] quantify this: albumin excreted by cells in such assays is "equivalent to a supplementation of 0.0003–0.0009%" of FBS, four to five orders of magnitude below the 2–10% used here. It cannot contribute.

Co-solvent. The three assays differ 30-fold in vehicle strength (Table 2). A co-solvent lowers every water-to-phase distribution ratio by approximately the same factor, and the predicted fraction is a ratio of sorptive terms, so the effect cancels except against the aqueous term. It is therefore material only where water carries a substantial share of the capacity — PFBA, where it changes the residual from 1.42× to 1.11×, and negligible elsewhere (PFNA, 9.33× to 9.31×). The 30-fold difference in vehicle is, counter-intuitively, of almost no consequence for this comparison.

6.7 A third parameterisation: measured biopartitioning

The design decision separating the Escher-school balances [12,14,15] from IV-MBM EQP v2.0 is the use of measured — or multi-descriptor predicted — membrane–water and protein–water partitioning in place of a single-descriptor regression, retaining a linear albumin term. Fischer et al. state the premise directly: distribution ratios derived "by log KOW correlation can lead to a high uncertainty and inaccuracy", whereas methods using several chemical properties "can substantially improve the quality of predicted partition constants" [12]. Armitage et al. report the same for their own model on ionisable organics [1]. Fischer et al. use polyparameter LFERs for neutral species, a 3D-QSAR for anionic albumin–water and COSMOmic for anionic lipid–water, preferring measured values where they exist. For PFAS measured values now exist, so the reconstruction below uses them: albumin from Fischer et al. [9], phospholipid membrane–water from Chen et al. [10]. PFBA is excluded (no measured albumin value), so comparisons are over nCF 5–8 and are directly comparable with the corresponding columns in §6.5.

Both measured series sit below the spLFER predictions — albumin by 2–7× and membrane lipid by 8–90× over nCF 5–8, with the membrane discrepancy growing systematically with chain length.

6.7.1 Parameter sensitivity within the Armitage structure

Substituting each measured series into IV-MBM EQP v2.0 while leaving its structure intact isolates the leverage of each coefficient.

Table 6. Coefficient substitutions within the Armitage structure, nCF 5–8.

parameterisation	bias	slope (decades/CF_2)
Armitage, as published	3.61×	+0.282
+ measured albumin only	8.40×	+0.160
+ measured membrane only	0.14×	-0.083
+ both measured	0.33×	-0.190
Qin, as published (saturable albumin)	7.13×	-0.077

Read as a sensitivity analysis, this table says that the measured membrane coefficient is where the chain-length drift lives: substituting it alone takes the slope from +0.282 to -0.083 and no other single substitution comes close.

It does **not**, however, describe the Escher-school design, and an earlier version of this section wrongly presented the last row as such. Within the Armitage structure, cell protein sorbs by octanol-equivalence, so it contributes almost nothing (§3.1); substituting a membrane coefficient 8–90× smaller than the spLFER's therefore collapses the cell sink almost to its water term, which is what produces the 0.33×. That is an artefact of mixing one model's structure with another's coefficients, not a result about either.

6.7.2 The Escher-school balance as published

Fischer et al.'s own form differs from the Armitage structure in four ways [12]. Their cell–water ratio is

> $D_{cell/w} = VF_{prot,cell} \cdot D_{BSA/w} + VF_{lip,cell} \cdot D_{lip/w} + VF_{w,cell}$

so (i) **one protein surrogate is used everywhere** — cell protein receives $D_{BSA/w}$, not an octanol equivalence, which is the difference that invalidates the reconstruction above; (ii) cell water is a phase of $D_{cell/w}$, unlike Qin's formulation; (iii) the normalising volume is V_{cell}, so the basis matches Armitage's and not Qin's; and (iv) there is no plastic term. The medium protein volume is also different in kind: it is total measured protein, not albumin. Fischer et al. measured 0.520% at 10% dialysed FBS, 3.0× the albumin volume the serum convention gives.

Fischer et al. also supply the cell composition these datasets most need — measured HepG2 (ARE-bla): 87.35% water, 9.46% protein, 3.19% lipid, 3.04 pg per cell, with all cellular lipid treated as membrane lipid. This matters because **Armitage's cell-line-specific entry is not a hepatic composition**. His AREc2 is AREc32, which is MCF7-derived (Cellosaurus CVCL_1D32), and carries 0.46% total lipid against the 3.19% measured for HepG2. Two of the three datasets modelled here use HepG2 and the third HepaRG, so the published parameterisation applies a breast-carcinoma composition throughout. (Armitage's other entry, for the PPAR*gamma* line, reproduces Fischer et al.'s measured HEK293 numbers to the digits shown, which is evidently its source.)

Table 7. The Fischer et al. form, parameterised on measured PFAS coefficients, nCF 5–8.

cells	medium protein	bias	slope (decades/CF_2)
measured HepG2 [12]	**measured total [12]**	**4.62×**	**-0.134**
measured HepG2 [12]	albumin volume only	14.19×	-0.067
Armitage AREc2 (MCF7)	measured total [12]	7.87×	-0.134
Armitage AREc2 (MCF7)	albumin volume only	22.95×	-0.078

Three results follow, and the first is a correction.

The level is not fixed by measured parameters. The faithful Escher-school balance over-predicts by 4.62×, against 3.61× for Armitage as published. The 0.33× of Table 6 was a structural artefact, and the claim that measured membrane partitioning moves the level toward unity does not survive. As it stands, **no parameterisation tested in this work fixes the level**, which strengthens rather than weakens the central negative result of §6.4.

The chain-length drift is removed. The slope falls from +0.282 to -0.134, and over nCF 5–7 — where three of the four compounds sit — it is -0.097. The reason is visible in the coefficients: Fischer's measured albumin series and Chen's measured membrane series rise with chain length at nearly the same rate, so their ratio, which is what the predicted fraction depends on (§6.5), is almost constant. Measured biopartitioning does solve the chain-length problem; it simply does not solve the level problem.

The two failures are cleanly separable. The slope in Table 7 is unchanged across a 6.9× change in cell lipid, a 3.3× change in cell volume and a 3× change in medium protein — it is -0.134 in both rows with measured medium protein and -0.07 to -0.08 in both without. Conversely the level moves by 5× across those same changes while the coefficients stay fixed. So:

> *the chain-length behaviour is a property of the partition coefficients; the level is a property of the volumes and compositions.*

Each error must be diagnosed with the corresponding class of parameter, and neither can be diagnosed from the other. Both improvements available from measurement work in the expected direction on the level — the measured hepatic composition gains 1.7× over the MCF7 one, and the measured medium protein gains 3.1× — and together they leave

4.62× standing.

Figure 2 places every parameterisation tested in this work on those two axes alongside the lipid series that drives the horizontal one. The discriminator for the slope is specifically the lipid coefficient: every variant whose lipid–water ratio is measured falls at or below zero, Qin et al.'s included since they take theirs from liposome measurements, and the only two that sit on the positive side are the two that predict the lipid term from log KOW.

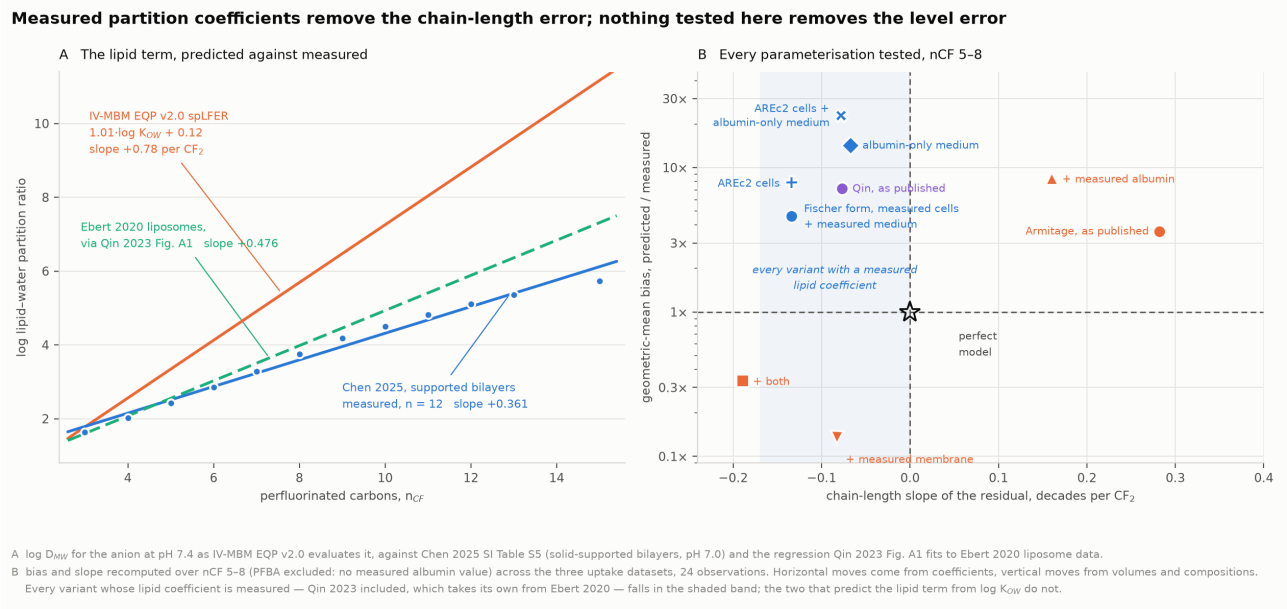


Figure 2. A, the lipid–water partition ratio against chain length. The orange line is log DMW for the anion at pH 7.4 as IV-MBM EQP v2.0 evaluates it (1.01·log KOW + 0.12, with the anion penalty dMW = 1.0); the blue points and line are Chen et al.'s measurements on solid-supported bilayers at pH 7.0 (SI Table S5, 12 perfluoroalkyl carboxylates); the dashed green line is the regression Qin et al. fit to Ebert et al.'s liposome data (Figure A1). The prediction rises more than twice as fast with chain length as either measurement. **B**, every parameterisation tested in this work placed on (chain-length slope, geometric-mean bias), recomputed over nCF 5–8 so both axes are on one footing; PFBA is excluded throughout because it has no measured albumin value. The star at (0, 1×) is a model with neither error. Substituting measured coefficients moves points horizontally; changing cell and medium volumes and compositions moves them vertically. Every variant whose lipid coefficient is measured falls in the shaded band — Qin et al.'s included, since they take theirs from liposome measurements — and the only two outside it are the two that predict the lipid term from log KOW. Nothing tested reaches the star.

Chen's coefficients carry the slope result in both Tables 6 and 7, so the basis on which they are reported has to be established. It can be. Chen et al. define KMW in their Supplementary equation S1 as

> $K_{MW} = (C_{spike} - C_{PFAS,W}) \cdot V_W / (C_{PFAS,W} \cdot M_{PL})$

where Cspike is the aqueous concentration in a lipid-free control at the end of the incubation, CPFAS,W the aqueous concentration in the sample, VW the water volume and MPL the quantity of phospholipid membrane in the tube; the authors note that a unit conversion is applied so that the reported coefficient is dimensionless. The measurement uses a commercial solid-supported lipid membrane kit in which the membrane is supplied as a stated *volume* — 0.067 to 2.166 µL of bilayer per tube in 500 µL of PBS — so the lipid quantity is natively volumetric and the mass and volume bases coincide at a phospholipid density of 1 g·cm-3. The reported log KMW is therefore a dimensionless concentration ratio on a volume-of-membrane to volume-of-water basis: the same basis as Armitage's DMW, and the same density convention the cell-volume conversion used here already adopts (10 ng per cell at 1 g·cm-3). Real phospholipid densities of 1.01–1.06 g·cm-3 bound the residual basis error at <=0.03 log units, which is two orders of magnitude smaller than the discrepancies in Tables 6 and 7. **The slope results therefore do not rest on an unverified basis.**

Two independent checks support this reading. The tabulated values reproduce the authors' own unweighted regression of log KMW on chain length — we obtain +0.361 (R^2 = 0.983) for the perfluoroalkyl carboxylates and +0.374 (R^2 = 0.985) for the sulfonates against their stated +0.361 ± 0.014 (R^2 = 0.982) and +0.374 ± 0.019 (R^2 = 0.983) — confirming that the series was transcribed correctly. And the absolute values sit in the published range for PFAS membrane–water partitioning measured by the same technique.

The slope comparison is anchored at source: the measured series rises +0.361 decades per CF_2 across the authors' full C4–C16 carboxylate range (+0.44 over the nCF 5–8 window used in Table 6), against +0.78 for the spLFER, which inherits +0.77 per CF_2 from the log K_{OW} series and multiplies it by 1.01. The spLFER over-steepens membrane partitioning by a factor of about 2.6 per CF_2 added; compounded over Chen's measured range this reaches three orders of magnitude, and the predicted-to-measured ratio runs from 0.7 at PFBA to 0.001 at PFDoA.

A third, fully independent series agrees. Qin et al. [3] take their lipid–water ratios from liposome measurements [20] and report the regression they used to fill in PFBA: slope **+0.476** per fluorinated carbon for the carboxylates. Their two sulfonate entries imply **+0.38**, against Chen's **+0.374** for the same head group. So two laboratories, two lipid preparations — phospholipid vesicles and solid-supported bilayers — and two analytical routes place the lipid–water chain-length slope between +0.36 and +0.48, while the spLFER inherits +0.77 from the log K_{OW} series and multiplies it by 1.01. **The over-steepening of the lipid term is not an artefact of any one measurement.** It is the single most consistently corroborated discrepancy in this reassessment, and it is the term that represents almost the whole of the Armitage cell at long chain length (Table 1).

One genuine caveat remains, and it is small. Chen et al. report K_{MW} at pH 7.0 while the balance is evaluated at pH 7.4, so a 0.4-unit extrapolation is implicit. The authors measured at pH 6 and pH 7 and present both, which bounds the pH sensitivity empirically, but they do not extend the series to 7.4.

The wider implication for the landscape is that the two failure modes documented here map onto two independent design axes. Measured biopartitioning addresses the level; a saturable albumin term addresses the chain-length behaviour. No published model currently occupies the corner that combines both with a protein-realistic cell.

6.8 How much agreement did the source models claim?

A model's own validation sets a floor on what an external benchmark can resolve, so it is worth asking what each model was shown to achieve before being applied here.

Qin et al. [3] measured cell–water distribution ratios for their own HEK293H line and compared them with the prediction from their Eq. 17. Reconstructing that comparison from their published composition and parameters reproduces their stated figures exactly:

Table 8. Qin's Eq. 17 against Qin's own measured $D_{cell/w}$.

	predicted	measured	*delta* log	predicted/measured	% of log value
PFBA	1.63	2.56	-0.93	0.12×	-36%
PFOA	3.69	3.27	+0.42	2.65×	+13%
PFHxS	3.54	3.23	+0.31	2.05×	+10%
PFOS	4.84	4.19	+0.65	4.49×	+16%

The paper describes this as agreement "within 15%" for PFOA, PFHxS and PFOS and "around 35% lower" for PFBA. The final column shows what those percentages are percentages of: the *logarithm* of the distribution ratio. On the quantity that enters the mass balance the same residuals are a 2.0–4.5× over-prediction for the three hydrophobic compounds and an 8.5× under-prediction for PFBA. This is not a criticism of the measurement or of the model's construction, and it does not affect the authors' conclusions, which concern the free concentration in a medium where albumin carries more than 98% of the capacity and the cell term is therefore nearly irrelevant. But it has two consequences for the present work.

First, the cell-term residual is **ordered by chain length**: within the carboxylates it gains +1.35 log units between nCF 3 and nCF 7, or +0.34 per CF_2. That is the same sign and nearly the same magnitude as the drift found above in the Armitage cell term (+0.38 per CF_2, Table 5) — in a model whose cell parameters are measured rather than predicted. The chain-length problem in the cell term is therefore not specific to the spLFER approach.

Second, a 2–4.5× discrepancy in $D_{cell/w}$ is **inside the envelope the model was published with**. An external benchmark cannot attribute a level error smaller than that to the model rather than to its inputs, which bounds how firmly the biases in Table 3 can be interpreted.

The same floor cannot be established for IV-MBM EQP v2.0 from the published record, because the cell-protein sorption parameter is not documented there at all (§3.1, §9.3).

6.9 A head-group test: the sulfonates

Everything above is a carboxylate result. Fragki et al. [7] also measured uptake for two perfluoroalkane sulfonates, PFHxS and PFOS, in the same wells, the same medium and the same protocol as their carboxylates, which makes the head-group contrast a within-study comparison with the study-level variables controlled. Qin et al. [3] measured every parameter for exactly those two compounds, so their limb needs no interpolation here — cleaner than for three of the five carboxylates.

Table 9. Fragki et al. [7] only, by head group. Geometric-mean bias.

	compounds	IV-MBM EQP v2.0	Qin et al.
carboxylates	PFOA, PFNA	6.66×	9.20×
sulfonates	**PFHxS, PFOS**	**0.98×**	7.38×

Armitage's over-prediction is head-group specific; Qin's is not. On the sulfonates IV-MBM EQP v2.0 is unbiased to 2%, against a factor of 6.7 on the carboxylates measured alongside them.

The per-compound measurements show why, and it is not a success:

	measured uptake	Armitage predicts
PFOA	1.03%	6.56%
PFNA	2.61%	18.21%
PFHxS	1.10%	0.94%
PFOS	6.62%	7.44%

PFHxS and PFOA are taken up almost identically — 1.10% against 1.03% — while the model separates them sevenfold, because their OPERA log KOW values differ by two log units (3.69 against 5.68). The agreement on the sulfonates is a coincidence of level between two compounds the model happens to place correctly, not evidence that the sulfonate case is better described. What the comparison actually shows is that **measured cellular uptake does not order these compounds the way log KOW does** — the same conclusion §6.7 reaches along the chain-length axis, reached here along the head-group axis with an independent set of compounds.

Two cautions. This rests on two compounds in one laboratory, so it is a test the published record barely supports; and PFHxS carries a log KOW disagreement of its own, 3.69 by OPERA against 5.29 by COSMOtherm (Qin Table 1), which moves the Armitage bias from 0.98× to 3.4× and which nothing here can adjudicate. The sulfonates are reported separately rather than pooled into the 29 observations precisely because head group would otherwise be confounded with study.

7. The ligand:protein regime

The argument of this section is not new. Henneberger et al. [21] established that for anionic chemicals "sorption to albumin cannot be predicted with a simple partition since saturation of albumin binding sites for these chemicals occurs at concentrations relevant for toxicity testing", as Proença et al. [17] summarise it, and the same review notes that organic acids appear to have only one binding site, which makes them saturate at relatively low concentrations. Qin et al. [3] carried the argument into QIVIVE: concentration-dependent protein binding makes the binding regime in bioassays differ substantially from that in plasma. Their full text states the mechanism directly: the gap between their nominal and free QIVIVE ratios arises "because protein binding is highly concentration-dependent with a high-affinity specific binding at low concentrations and lower non-specific binding at higher concentrations. Thus, the low concentrations in human plasma are strongly bound to proteins, while the high concentration needed to trigger effects in in vitro bioassays have a higher fraction unbound." What is added here is the magnitude of the gap for three datasets neither group analysed, and the demonstration that it is large enough to invalidate the linearity assumption within a single dose series. Figure 3 quantifies it.

Albumin offers approximately one high-affinity site per molecule for these compounds (§3.2). The two 10% FBS assays contain 36.1 µM albumin and the 2% FBS assay 7.2 µM, against nominal doses of 6.25–100 µM. The resulting nominal

ligand:albumin ratios are 0.28–2.77 and 0.87–13.85 mol/mol respectively. Fischer et al. [9] worked at <= 0.004 mol/mol and estimate the ratio in human serum at ~ 0.0005. **The assays therefore operate 5,500–27,700× above the physiological regime.**

The consequence for the models is direct. Solved self-consistently within the wells, albumin site occupancy runs from 20% to 99% across the dose series, and the apparent albumin distribution ratio falls to between 0.11 and 0.07 of its trace-limit value — a sevenfold change in a quantity that a linear model treats as a constant, within a single experiment (Fig. 3B).

This is why the Qin formulation requires its saturable term and why a linear spLFER treatment cannot be correct here, whatever its merits for the neutral organics it was calibrated on.

It also bounds what the exercise can deliver. Even an exactly correct model of these wells would describe a binding regime that does not occur in vivo. Improving the in vitro mass balance improves the description of the experiment; it does not, on its own, improve the extrapolation, because the experiment is in the wrong regime.

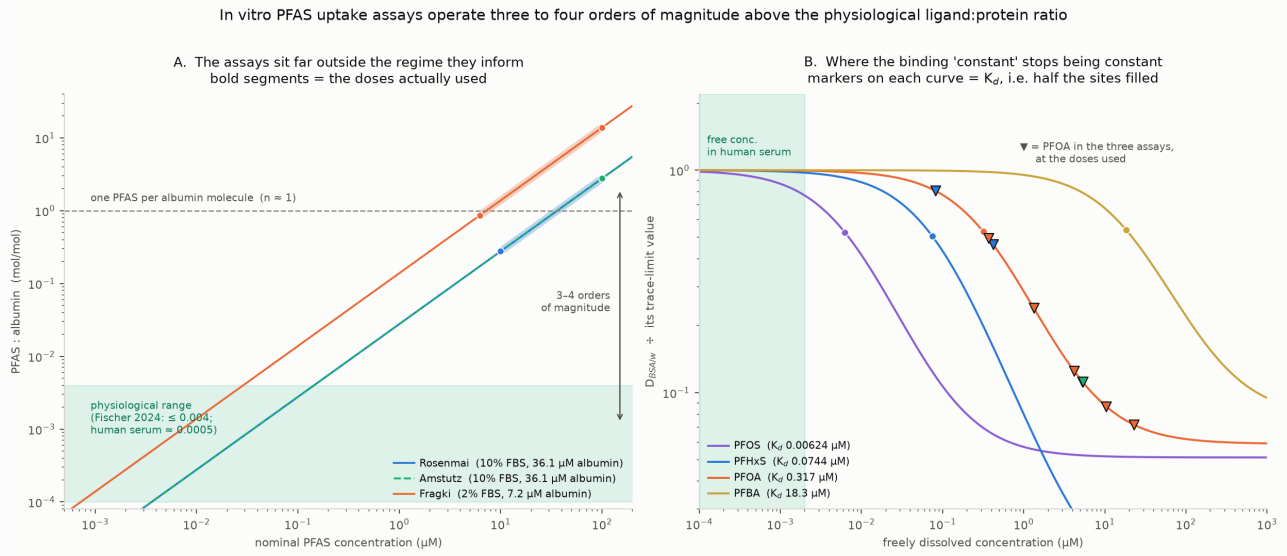


Figure 3. A, the nominal ligand:albumin molar ratio against dose for the three assays, with the physiological band marked. Albumin offers approximately one high-affinity site per molecule for these compounds (§3.2), so a ratio of 1 means the sites are nominally filled. All three assays run three to four orders of magnitude above the band. **B**, the consequence for the models: the apparent albumin distribution ratio, relative to its trace limit, as a function of freely dissolved concentration, with the positions the three assays occupy marked. Within a single dose series the quantity a linear model treats as a constant falls to between 0.11 and 0.07 of its trace value.

8. A positive control: the same formalism in serum

The results above could be read as indicting the mass-balance approach itself. A control argues otherwise.

Fischer et al. [9] measured, by one method in one laboratory, PFAS distribution ratios to purified human serum albumin, to *gamma*-globulin, and to whole human serum, at a ligand:protein ratio of <= 0.004. The components and the whole matrix are separate experiments, so predicting the latter from the former is an out-of-sample test of the identical accounting used above.

For the 26 PFAS with all three measurements, the balance evaluated at population-average serum composition and with no fitted parameter gives a geometric-mean bias of 1.00× and a geometric-mean absolute error of 1.55× (RMSE 0.262 log units; R^2 = 0.68 about the 1:1 line) for a quantity spanning nearly two orders of magnitude. Allowing the two protein concentrations to float returns 39.5 and 38.6 g/L against population values of 41 and 31 g/L and does not reduce the RMSE at all — the model is not being supported by its parameters. Globulin earns its term only above the ratio threshold the authors themselves propose. Adding a phospholipid sink improves nothing.

The accounting is therefore sound, and the failure is specific to the in vitro system — which has more phases, a vessel-wall term that is poorly constrained, cell compositions the studies themselves did not measure, and a

ligand:protein regime in which the central linearity assumption does not hold.

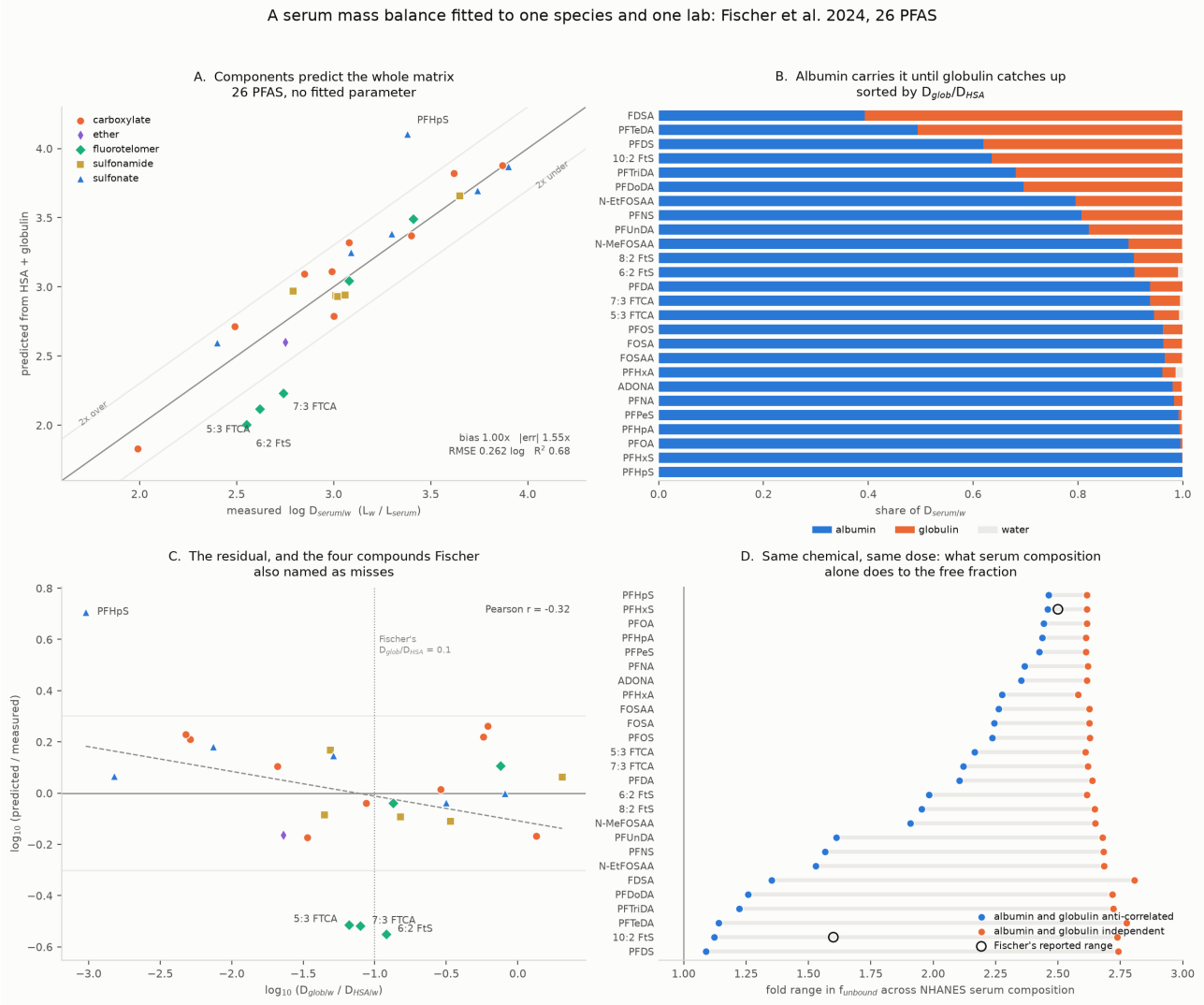


Figure 4. The same mass-balance formalism applied to human serum, as a positive control. Predicted against measured whole-serum–water distribution for the 26 PFAS in Fischer et al.'s Table S2, using their measured albumin coefficients, physiological albumin and globulin concentrations, and no fitted parameter. Geometric-mean bias 1.00×, geometric-mean |error| 1.55×, $R^2 = 0.68$. The accounting is sound; the failure documented in §6 is specific to the in vitro system.

9. Discussion

9.1 Implications for QIVIVE

Three findings bear on the use of these assays for reverse dosimetry.

First, the models over-predict cellular uptake by factors of 2–9 on data whose own inter-study spread is a factor of 17. Neither model is currently accurate enough to carry a QIVIVE calculation without an uncertainty statement of comparable size.

Second, the residual is not attributable to any parameter for which a better measurement exists. Substituting the best available value for albumin, for cell partitioning and for cell protein each worsens the fit. The only parameter that improves it is the one nobody measured.

Third, and most consequentially, the regime gap of §7 means that improving the in vitro balance does not translate directly into a better oral equivalent dose. Qin et al. [3] reached the same conclusion from the opposite direction,

showing that free-concentration-based QIVIVE ratios exceed nominal-based ones precisely because of protein-content differences between blood and bioassays. Our contribution is to show that the gap is large enough — three to four orders of magnitude in ligand:protein ratio — that the saturation state of albumin differs qualitatively, not merely quantitatively, between the two settings.

9.2 Reporting recommendations

Several of the difficulties encountered here are avoidable at no experimental cost.

1. **Report the exposure medium composition, not the culture medium composition.** The single most consequential error in this work arose from a five-fold difference between the two in one study, stated in one sentence of its methods and absent from derived descriptions.
2. **Report the working volume.** One of the three studies does not, leaving a parameter that enters the balance linearly to be assumed.
3. **Report the cell volume, or cells per well together with protein per cell.** Cell volume is the only parameter that improved the fit here and the only one none of the studies measured.
4. **Report the ligand:protein molar ratio.** It determines whether a measured binding constant is in the trace regime and therefore whether it can be transferred to another setting at all.
5. **State the normalisation basis** — administered or recovered — since the two differ by whatever was lost to the vessel.
6. **Report the mass balance where it was measured.** Two of the three studies took a t = 0 medium sample explicitly for this purpose; neither reports the result. Doing so would constrain the vessel-wall term directly rather than by difference.

9.3 Limitations of this reassessment

The Armitage implementation depends on one parameter — the sorptive capacity assigned to bulk cell protein — that is not documented in the published record (§3.1). We used the `httk` convention. Having read both papers, we can now bound what this costs. At the `httk` value the protein phase carries 0.05–0.25% of the cell term, so no result here is sensitive to the number itself; what the papers do not settle is whether the authors' own tool makes the same octanol-equivalence *choice*, and that choice does have leverage (§6.4). Deciding it would require the Excel/VBA implementation, which is available from the authors on request. The structural argument is unaffected either way: a cell whose protein term is small is represented as lipid, and a lipid representation carries too steep a chain-length dependence — which three independent measured lipid series now corroborate (§6.7).

Qin's parameters exist for four compounds; two of the five modelled here are interpolated and one extrapolated. The extrapolated compound, PFNA, is among the worst-fitting. The authors scope their own claim narrowly — they demonstrated the model's validity "only for four anionic PFAS, one type of bioassay medium and one cell line" — while arguing that prior work on other organic anions supports transfer to other setups "if the lipid and protein content has been quantified". Applying the model to three other datasets with quantified composition is therefore the use the authors anticipate; extending it along the chain-length axis to compounds they did not measure is not, and the interpolation used here for PFHxA and PFHpA and the extrapolation for PFNA have no published warrant beyond the lipid term.

The measured-membrane result of §6.7 rests on a coefficient reported at pH 7.0 and applied here at pH 7.4; the reporting basis itself is resolved (§6.7) but the 0.4-unit pH extrapolation is not.

The empirical basis is almost entirely perfluoroalkyl carboxylates. The head-group test of §6.9 rests on two sulfonates in one laboratory, and it already shows IV-MBM EQP v2.0 behaving differently there (unbiased rather than 6.7× high). Whether the chain-length result of §6.5 and §6.7 holds for sulfonates cannot be decided from the published uptake data, because no study reports a sulfonate chain-length series.

The uptake datasets are small, mutually inconsistent, and partly not on a common normalisation basis. Rosenmai's working volume is assumed. Conclusions about absolute bias are correspondingly weaker than conclusions about *direction* and about *structure*, which is why we have emphasised the latter.

Finally, of the fourteen models Proença et al. [17] catalogue, three are evaluated here — the two under assessment and a reconstruction of Fischer et al. [12]. The remaining eleven are not, and the comparison here should not be read as a ranking of the field. The Fischer reconstruction in particular is built from that paper's published equations and its Table 2

system parameters, not from the authors' own implementation, and substitutes measured PFAS coefficients for the PP-LFER, 3D-QSAR and COSMOmic predictions they would use; it is the design tested on the best available PFAS data, which is not the same thing as the model as its authors would run it.

10. Conclusions

1. Both mass-balance models over-predict PFAS cellular uptake in human liver cell lines, by 3.3× (IV-MBM EQP v2.0) and 6.4× (Qin et al.) in geometric mean, once exposure conditions and uptake values are read from primary sources.
2. Exposure media differ between the three available datasets in ways that dominate the model comparison and are not recoverable from secondary descriptions. One assay was dosed at one-fifth the serum concentration generally assumed for it.
3. Every substitution of a best-available measured parameter worsens the fit. Albumin is the most influential term in the balance and is not the source of the error.
4. The residual separates into a level component and a chain-length component, and the two are governed by different classes of parameter: the chain-length behaviour by the partition coefficients, the level by the volumes and compositions. Each is invariant to large changes in the other, so neither error can be diagnosed with the wrong class of parameter.
5. Measured biopartitioning removes the chain-length error and not the level error. A faithful reconstruction of the measured-parameter linear design is flat over nCF 5–7 and still over-predicts by 4.6×. The cause of the chain-length error is specifically that single-descriptor prediction over-steepens the lipid term: three independent measured series place the lipid–water chain-length slope at +0.36 to +0.48 decades per CF_2 against +0.78 predicted.
6. All three assays operate three to four orders of magnitude above the physiological ligand:protein ratio, with albumin between 20% and 99% saturated. The linearity assumption underlying spLFER-based balances does not hold in this regime.
7. The same formalism applied to human serum reproduces measured whole-serum partitioning for 26 PFAS with no fitted parameter. The accounting is sound; the in vitro system is the problem.
8. Consequently, better in vitro dosimetry yields a better description of the experiment, but not by itself a better oral equivalent dose. Closing that gap requires assays run nearer the physiological binding regime, not better models of assays that are not.

Data and code availability

model implementations (§3.1, §3.2)	`tk_review/scripts/mbm_published_equations.py`
exposure conditions, source-verified (§4.2)	`tk_review/scripts/mbm_medium_composition.py`
chain-length decomposition (§6.5)	`tk_review/scripts/mbm_chain_length_residual.py`
parameter substitutions (§6.4)	`mbm_fischer_albumin.py`, `mbm_cell_term.py`, `mbm_cell_protein.py`
coefficient substitutions, Armitage structure (§6.7.1)	`tk_review/scripts/mbm_measured_parameter_variant.py`
the Fischer et al. form (§6.7.2)	`tk_review/scripts/mbm_fischer_escher_balance.py`
Qin's model against Qin's measurement (§6.8)	`tk_review/scripts/mbm_qin_internal_check.py`
IV-MBM v1.0 against v2.0 (§3.1)	`tk_review/scripts/mbm_armitage_v1_vs_v2.py`
serum control (§8)	`tk_review/scripts/mbm_fischer_serum.py`
Figure 1	`tk_review/scripts/make_mbm_correction_figure.py`
Figure 2	`tk_review/scripts/make_mbm_separation_figure.py`
Figure 3	`tk_review/scripts/make_mbm_regime_figure.py`

model implementations (§3.1, §3.2)	`tk_review/scripts/mbm_published_equations.py`
Figure 4	`tk_review/scripts/make_mbm_fischer_serum_figure.py`
primary sources held locally	`literature/fulltext/`

Extended derivations and the full approximation inventory are in `report/MBM_PRIMER.md`; a non-technical exposition is in `report/MBM_UNDERSTANDING.md`; the detailed working memo, including material not carried into this manuscript, is `report/MBM_Qin_vs_Armitage.md`.

References

1. Armitage JM, Sangion A, Parmar R, Looky AB, Arnot JA. Update and evaluation of a high-throughput in vitro mass balance distribution model: IV-MBM EQP v2.0. *Toxics* 2021;9:315. doi:10.3390/toxics9110315
2. Armitage JM, Wania F, Arnot JA. Application of mass balance models and the chemical activity concept to facilitate the use of in vitro toxicity data for risk assessment. *Environ Sci Technol* 2014;48(16):9770–9779. doi:10.1021/es501955g
3. Qin W, Henneberger L, Huchthausen J, König M, Escher BI. Role of bioavailability and protein binding of four anionic perfluoroalkyl substances in cell-based bioassays for quantitative in vitro to in vivo extrapolations. *Environ Int* 2023;173:107857. doi:10.1016/j.envint.2023.107857
4. Qin W, Escher BI, Huchthausen J, Fu Q, Henneberger L. Species difference? Bovine, trout, and human plasma protein binding of per- and polyfluoroalkyl substances. *Environ Sci Technol* 2024;58(23):9954–9966. doi:10.1021/acs.est.3c10824
5. Rosenmai AK, Ahrens L, le Godec T, Lundqvist J, Oskarsson A. Relationship between peroxisome proliferator-activated receptor alpha activity and cellular concentration of 14 perfluoroalkyl substances in HepG2 cells. *J Appl Toxicol* 2018;38:219–226. doi:10.1002/jat.3515
6. Amstutz VH, Cengo A, Gehres F, Sijm DTHM, Vrolijk MF. Investigating the cytotoxicity of per- and polyfluoroalkyl substances in HepG2 cells: a structure-activity relationship approach. *Toxicology* 2022;480:153312. doi:10.1016/j.tox.2022.153312
7. Fragki S, Louisse J, Bokkers B, Luijten M, Peijnenburg A, Rijkers D, Piersma AH, Zeilmaker MJ. New approach methodologies: a quantitative in vitro to in vivo extrapolation case study with PFASs. *Food Chem Toxicol* 2022;172:113559. doi:10.1016/j.fct.2022.113559
8. Louisse J, Fragki S, Rijkers D, et al. Determination of in vitro hepatotoxic potencies of a series of perfluoroalkyl substances (PFASs) based on gene expression changes in HepaRG liver cells. *Arch Toxicol* 2023;97:1113–1131. doi:10.1007/s00204-023-03450-2
9. Fischer FC, et al. *Environ Sci Technol* 2024;58(2):1055–1063. doi:10.1021/acs.est.3c07415
10. Chen R, Muensterman D, Field J, Ng C. Deriving membrane-water and protein-water partition coefficients from in vitro experiments for per- and polyfluoroalkyl substances (PFAS). *Environ Sci Technol* 2025;59(1):82–91. doi:10.1021/acs.est.4c06734
11. Ryu S, Burchett W, Zhang S, Jia X, Modaresi SMS, Agudelo Areiza J, Rodrigues D, Zhu H, Sunderland EM, Fischer FC, Slitt AL. Unbound fractions of PFAS in human and rodent tissues: rat liver a suitable proxy for evaluating emerging PFAS? *Environ Sci Technol* 2024;58(33):14641–14650. doi:10.1021/acs.est.4c04050
12. Fischer FC, Henneberger L, König M, Bittermann K, Linden L, Goss K-U, Escher BI. Modeling exposure in the Tox21 in vitro bioassays. *Chem Res Toxicol* 2017;30(5):1197–1208. doi:10.1021/acs.chemrestox.7b00023
13. Fischer FC, Cirpka OA, Goss K-U, Henneberger L, Escher BI. Application of experimental polystyrene partition constants and diffusion coefficients to predict the sorption of neutral organic chemicals to multiwell plates. *Environ Sci Technol* 2018;52:13511–13522.
14. Henneberger L, Mühlenbrink M, Fischer FC, Escher BI. C18-coated solid-phase microextraction fibers for the quantification of partitioning of organic acids to proteins, lipids, and cells. *Chem Res Toxicol* 2019;32:168–178.
15. Huchthausen J, Mühlenbrink M, König M, Escher BI, Henneberger L. Experimental exposure assessment of ionizable organic chemicals in in vitro cell-based bioassays. *Chem Res Toxicol* 2020;33:1845–1854.

16. Endo S, Brown TN, Goss K-U. General model for estimating partition coefficients to organisms and their tissues using the biological compositions and polyparameter linear free energy relationships. *Environ Sci Technol* 2013;47:6630–6639.

17. Proença S, Escher BI, Fischer FC, Fisher C, Grégoire S, Hewitt NJ, Nicol B, Paini A, Kramer NI. Effective exposure of chemicals in in vitro cell systems: a review of chemical distribution models. *Toxicol In Vitro* 2021;73:105133. doi:10.1016/j.tiv.2021.105133

18. Stadnicka-Michalak J, Tanneberger K, Schirmer K, Ashauer R. Measured and modeled toxicokinetics in cultured fish cells and application to in vitro–in vivo toxicity extrapolation. *PLoS One* 2014;9(3):e92303. doi:10.1371/journal.pone.0092303

19. Paini A, Sala Benito JV, Bessems J, Worth AP. From in vitro to in vivo: integration of the virtual cell based assay with physiologically based kinetic modelling. *Toxicol In Vitro* 2017;45(Pt 2):241–248. doi:10.1016/j.tiv.2017.06.015

20. Ebert A, Allendorf F, Berger U, Goss K-U, Ulrich N. Membrane/water partitioning and permeabilities of perfluoroalkyl acids and four of their alternatives and the effects on toxicokinetic behavior. *Environ Sci Technol* 2020;54(8):5051–5061. doi:10.1021/acs.est.0c00157 — Source of the liposome–water ratios used by ref. [3]; cited here via [3], whose Table 2 and Figure A1 regression are the values used.

21. Henneberger L, Mühlenbrink M, Heinrich DJ, Teixeira A, Nicol B, Escher BI. Experimental validation of mass balance models for in vitro cell-based bioassays. *Environ Sci Technol* 2020;54(2):1120–1127. doi:10.1021/acs.est.9b06144 — Cited via ref. [17] for the saturation of albumin binding by anionic chemicals at toxicologically relevant concentrations; not independently retrieved.

Provenance of references. Read at source, including supplementary material where cited: 1, 2, 3, 5, 6, 7, 8, 9, 10, 12, 17. Bibliographic record verified: 4, 11, 18, 19. Cited via ref. [3], which is the source of the values used: 20. Cited via ref. [17] for a statement attributed to it, not independently retrieved: 21. Cited as methodological context from the reference list of ref. 1, not independently retrieved: 13, 14, 15, 16.

Both models under assessment have now been read at source. For ref. [3] the full text supplied the balance equations, the cell composition and the authors' own cell-term validation, all used in §3.2 and §6.8. For ref. [1] the supplementary information supplied every parameter except the cell-protein sorption term; ref. [2] establishes that no such term existed in v1.0 and that the 3.5% convention is not in the published record (§3.1, §9.3). Only the main text of ref. [2] was read; its supplementary sections S1–S9 were not retrieved, and the statements made here rest on its Eq. 1 and the surrounding model description, which are in the main text. Ref. [12] was read in full; only its main text was retrieved, and the cell and medium compositions used here are from its Table 2, not its Tables S1–S2, which carry the full set of cell lines and assays. The AREc32 provenance is from Cellosaurus (CVCL_1D32) and Wang, Hayes & Wolf 2006, not from ref. [1], which does not state the line's derivation.